MSG No. 97.156 I_c422 [Type IV, tetragonal]

Table 1: Wyckoff site: `4a`, site symmetry: `422`

No.	position	mapping
1	$[0, 0, 0]$	`[1,2,3,4,5,6,7,8]`
2	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[9,10,11,12,13,14,15,16]`
3	$\left[0, 0, \frac{1}{2}\right]$	`[17,18,19,20,21,22,23,24]`
4	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[25,26,27,28,29,30,31,32]`

Table 2: Wyckoff site: `4b`, site symmetry: `42'2'`

No.	position	mapping
1	$\left[0, 0, \frac{1}{4}\right]$	`[1,2,3,6,20,21,23,24]`
2	$\left[0, 0, \frac{3}{4}\right]$	`[4,5,7,8,17,18,19,22]`
3	$\left[\frac{1}{2}, \frac{1}{2}, \frac{3}{4}\right]$	`[9,10,11,14,28,29,31,32]`
4	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{4}\right]$	`[12,13,15,16,25,26,27,30]`

Table 3: Wyckoff site: `4c`, site symmetry: `4'22'`

No.	position	mapping
1	$\left[0, \frac{1}{2}, 0\right]$	`[1,4,5,6,26,27,31,32]`
2	$\left[\frac{1}{2}, 0, 0\right]$	`[2,3,7,8,25,28,29,30]`
3	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	`[9,12,13,14,18,19,23,24]`
4	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	`[10,11,15,16,17,20,21,22]`

Table 4: Wyckoff site: `4d`, site symmetry: `4'2'2`

No.	position	mapping
1	$\left[0, \frac{1}{2}, \frac{1}{4}\right]$	`[1,6,15,16,20,21,26,27]`
2	$\left[\frac{1}{2}, 0, \frac{1}{4}\right]$	`[2,3,12,13,23,24,25,30]`
3	$\left[0, \frac{1}{2}, \frac{3}{4}\right]$	`[4,5,10,11,17,22,31,32]`
4	$\left[\frac{1}{2}, 0, \frac{3}{4}\right]$	`[7,8,9,14,18,19,28,29]`

Table 5: Wyckoff site: `8e`, site symmetry: `2.22`

No.	position	mapping
1	$\left[\frac{1}{4}, \frac{1}{4}, 0\right]$	`[1,7,30,32]`
2	$\left[\frac{3}{4}, \frac{1}{4}, 0\right]$	`[2,5,27,28]`
3	$\left[\frac{1}{4}, \frac{3}{4}, 0\right]$	`[3,4,26,29]`
4	$\left[\frac{3}{4}, \frac{3}{4}, 0\right]$	`[6,8,25,31]`
5	$\left[\frac{3}{4}, \frac{3}{4}, \frac{1}{2}\right]$	`[9,15,22,24]`
6	$\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{2}\right]$	`[10,13,19,20]`
7	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{2}\right]$	`[11,12,18,21]`

8	$\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{2}\right]$	[14,16,17,23]

Table 6: Wyckoff site: 8f, site symmetry: 2.22

No.	position	mapping
1	$\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}\right]$	[1,16,23,30]
2	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{4}\right]$	[2,12,21,27]
3	$\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{4}\right]$	[3,13,20,26]
4	$\left[\frac{1}{4}, \frac{3}{4}, \frac{3}{4}\right]$	[4,10,19,29]
5	$\left[\frac{3}{4}, \frac{1}{4}, \frac{3}{4}\right]$	[5,11,18,28]
6	$\left[\frac{3}{4}, \frac{3}{4}, \frac{1}{4}\right]$	[6,15,24,25]
7	$\left[\frac{1}{4}, \frac{1}{4}, \frac{3}{4}\right]$	[7,14,17,32]
8	$\left[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}\right]$	[8,9,22,31]

Table 7: Wyckoff site: 8g, site symmetry: 4..

No.	position	mapping
1	$[0, 0, z]$	[1,2,3,6]
2	$[0, 0, -z]$	[4,5,7,8]
3	$\left[\frac{1}{2}, \frac{1}{2}, z+\frac{1}{2}\right]$	[9,10,11,14]
4	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}-z\right]$	[12,13,15,16]
5	$\left[0, 0, z+\frac{1}{2}\right]$	[17,18,19,22]
6	$\left[0, 0, \frac{1}{2}-z\right]$	[20,21,23,24]
7	$\left[\frac{1}{2}, \frac{1}{2}, z\right]$	[25,26,27,30]
8	$\left[\frac{1}{2}, \frac{1}{2}, -z\right]$	[28,29,31,32]

Table 8: Wyckoff site: 8h, site symmetry: 4'..

No.	position	mapping
1	$\left[0, \frac{1}{2}, z\right]$	[1,6,26,27]
2	$\left[\frac{1}{2}, 0, z\right]$	[2,3,25,30]
3	$\left[0, \frac{1}{2}, -z\right]$	[4,5,31,32]
4	$\left[\frac{1}{2}, 0, -z\right]$	[7,8,28,29]
5	$\left[\frac{1}{2}, 0, z+\frac{1}{2}\right]$	[9,14,18,19]
6	$\left[0, \frac{1}{2}, z+\frac{1}{2}\right]$	[10,11,17,22]
7	$\left[\frac{1}{2}, 0, \frac{1}{2}-z\right]$	[12,13,23,24]
8	$\left[0, \frac{1}{2}, \frac{1}{2}-z\right]$	[15,16,20,21]

Table 9: Wyckoff site: 16i, site symmetry: 2'..

No.	position	mapping
1	$\left[\frac{3}{4}, \frac{1}{4}, z\right]$	[1,30]
2	$\left[\frac{3}{4}, \frac{3}{4}, z\right]$	[2,27]
3	$\left[\frac{1}{4}, \frac{1}{4}, z\right]$	[3,26]

continued ...

Table 9

No.	position	mapping
4	$\left[\frac{3}{4},\ \frac{3}{4},\ -z\right]$	[4,29]
5	$\left[\frac{1}{4},\ \frac{1}{4},\ -z\right]$	[5,28]
6	$\left[\frac{1}{4},\ \frac{3}{4},\ z\right]$	[6,25]
7	$\left[\frac{1}{4},\ \frac{3}{4},\ -z\right]$	[7,32]
8	$\left[\frac{3}{4},\ \frac{1}{4},\ -z\right]$	[8,31]
9	$\left[\frac{1}{4},\ \frac{3}{4},\ z+\frac{1}{2}\right]$	[9,22]
10	$\left[\frac{1}{4},\ \frac{1}{4},\ z+\frac{1}{2}\right]$	[10,19]
11	$\left[\frac{3}{4},\ \frac{3}{4},\ z+\frac{1}{2}\right]$	[11,18]
12	$\left[\frac{1}{4},\ \frac{1}{4},\ \frac{1}{2}-z\right]$	[12,21]
13	$\left[\frac{3}{4},\ \frac{3}{4},\ \frac{1}{2}-z\right]$	[13,20]
14	$\left[\frac{3}{4},\ \frac{1}{4},\ z+\frac{1}{2}\right]$	[14,17]
15	$\left[\frac{3}{4},\ \frac{1}{4},\ \frac{1}{2}-z\right]$	[15,24]
16	$\left[\frac{1}{4},\ \frac{3}{4},\ \frac{1}{2}-z\right]$	[16,23]

Table 10: Wyckoff site: 16j, site symmetry: .2.

No.	position	mapping
1	$[0,\ y,\ 0]$	[1,5]
2	$[-y,\ 0,\ 0]$	[2,8]
3	$[y,\ 0,\ 0]$	[3,7]
4	$[0,\ -y,\ 0]$	[4,6]
5	$\left[\frac{1}{2},\ y+\frac{1}{2},\ \frac{1}{2}\right]$	[9,13]
6	$\left[\frac{1}{2}-y,\ \frac{1}{2},\ \frac{1}{2}\right]$	[10,16]
7	$\left[y+\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}\right]$	[11,15]
8	$\left[\frac{1}{2},\ \frac{1}{2}-y,\ \frac{1}{2}\right]$	[12,14]
9	$\left[0,\ y,\ \frac{1}{2}\right]$	[17,21]
10	$\left[-y,\ 0,\ \frac{1}{2}\right]$	[18,24]
11	$\left[y,\ 0,\ \frac{1}{2}\right]$	[19,23]
12	$\left[0,\ -y,\ \frac{1}{2}\right]$	[20,22]
13	$\left[\frac{1}{2},\ y+\frac{1}{2},\ 0\right]$	[25,29]
14	$\left[\frac{1}{2}-y,\ \frac{1}{2},\ 0\right]$	[26,32]
15	$\left[y+\frac{1}{2},\ \frac{1}{2},\ 0\right]$	[27,31]
16	$\left[\frac{1}{2},\ \frac{1}{2}-y,\ 0\right]$	[28,30]

Table 11: Wyckoff site: 16k, site symmetry: .2'.

No.	position	mapping
1	$\left[0,\ y,\ \frac{1}{4}\right]$	[1,21]
2	$\left[-y,\ 0,\ \frac{1}{4}\right]$	[2,24]
3	$\left[y,\ 0,\ \frac{1}{4}\right]$	[3,23]
4	$\left[0,\ -y,\ \frac{3}{4}\right]$	[4,22]
5	$\left[0,\ y,\ \frac{3}{4}\right]$	[5,17]
6	$\left[0,\ -y,\ \frac{1}{4}\right]$	[6,20]
7	$\left[y,\ 0,\ \frac{3}{4}\right]$	[7,19]

continued ...

Table 11

No.	position	mapping
8	$\left[-y,\ 0,\ \frac{3}{4}\right]$	[8,18]
9	$\left[\frac{1}{2},\ y+\frac{1}{2},\ \frac{3}{4}\right]$	[9,29]
10	$\left[\frac{1}{2}-y,\ \frac{1}{2},\ \frac{3}{4}\right]$	[10,32]
11	$\left[y+\frac{1}{2},\ \frac{1}{2},\ \frac{3}{4}\right]$	[11,31]
12	$\left[\frac{1}{2},\ \frac{1}{2}-y,\ \frac{1}{4}\right]$	[12,30]
13	$\left[\frac{1}{2},\ y+\frac{1}{2},\ \frac{1}{4}\right]$	[13,25]
14	$\left[\frac{1}{2},\ \frac{1}{2}-y,\ \frac{3}{4}\right]$	[14,28]
15	$\left[y+\frac{1}{2},\ \frac{1}{2},\ \frac{1}{4}\right]$	[15,27]
16	$\left[\frac{1}{2}-y,\ \frac{1}{2},\ \frac{1}{4}\right]$	[16,26]

Table 12: Wyckoff site: 16l, site symmetry: ..2

No.	position	mapping
1	$[x,\ x,\ 0]$	[1,7]
2	$[-x,\ x,\ 0]$	[2,5]
3	$[x,\ -x,\ 0]$	[3,4]
4	$[-x,\ -x,\ 0]$	[6,8]
5	$\left[x+\frac{1}{2},\ x+\frac{1}{2},\ \frac{1}{2}\right]$	[9,15]
6	$\left[\frac{1}{2}-x,\ x+\frac{1}{2},\ \frac{1}{2}\right]$	[10,13]
7	$\left[x+\frac{1}{2},\ \frac{1}{2}-x,\ \frac{1}{2}\right]$	[11,12]
8	$\left[\frac{1}{2}-x,\ \frac{1}{2}-x,\ \frac{1}{2}\right]$	[14,16]
9	$\left[x,\ x,\ \frac{1}{2}\right]$	[17,23]
10	$\left[-x,\ x,\ \frac{1}{2}\right]$	[18,21]
11	$\left[x,\ -x,\ \frac{1}{2}\right]$	[19,20]
12	$\left[-x,\ -x,\ \frac{1}{2}\right]$	[22,24]
13	$\left[x+\frac{1}{2},\ x+\frac{1}{2},\ 0\right]$	[25,31]
14	$\left[\frac{1}{2}-x,\ x+\frac{1}{2},\ 0\right]$	[26,29]
15	$\left[x+\frac{1}{2},\ \frac{1}{2}-x,\ 0\right]$	[27,28]
16	$\left[\frac{1}{2}-x,\ \frac{1}{2}-x,\ 0\right]$	[30,32]

Table 13: Wyckoff site: 16m, site symmetry: ..2

No.	position	mapping
1	$\left[x,\ x+\frac{1}{2},\ \frac{1}{4}\right]$	[1,15]
2	$\left[\frac{1}{2}-x,\ x,\ \frac{1}{4}\right]$	[2,13]
3	$\left[x+\frac{1}{2},\ -x,\ \frac{1}{4}\right]$	[3,12]
4	$\left[x,\ \frac{1}{2}-x,\ \frac{3}{4}\right]$	[4,11]
5	$\left[-x,\ x+\frac{1}{2},\ \frac{3}{4}\right]$	[5,10]
6	$\left[-x,\ \frac{1}{2}-x,\ \frac{1}{4}\right]$	[6,16]
7	$\left[x+\frac{1}{2},\ x,\ \frac{3}{4}\right]$	[7,9]
8	$\left[\frac{1}{2}-x,\ -x,\ \frac{3}{4}\right]$	[8,14]
9	$\left[x,\ x+\frac{1}{2},\ \frac{3}{4}\right]$	[17,31]
10	$\left[\frac{1}{2}-x,\ x,\ \frac{3}{4}\right]$	[18,29]
11	$\left[x+\frac{1}{2},\ -x,\ \frac{3}{4}\right]$	[19,28]

continued ...

Table 13

No.	position	mapping
12	$\left[x,\ \frac{1}{2}-x,\ \frac{1}{4}\right]$	[20,27]
13	$\left[-x,\ x+\frac{1}{2},\ \frac{1}{4}\right]$	[21,26]
14	$\left[-x,\ \frac{1}{2}-x,\ \frac{3}{4}\right]$	[22,32]
15	$\left[x+\frac{1}{2},\ x,\ \frac{1}{4}\right]$	[23,25]
16	$\left[\frac{1}{2}-x,\ -x,\ \frac{1}{4}\right]$	[24,30]

Table 14: Wyckoff site: 16n, site symmetry: ..2'

No.	position	mapping
1	$\left[x,\ x,\ \frac{1}{4}\right]$	[1,23]
2	$\left[-x,\ x,\ \frac{1}{4}\right]$	[2,21]
3	$\left[x,\ -x,\ \frac{1}{4}\right]$	[3,20]
4	$\left[x,\ -x,\ \frac{3}{4}\right]$	[4,19]
5	$\left[-x,\ x,\ \frac{3}{4}\right]$	[5,18]
6	$\left[-x,\ -x,\ \frac{1}{4}\right]$	[6,24]
7	$\left[x,\ x,\ \frac{3}{4}\right]$	[7,17]
8	$\left[-x,\ -x,\ \frac{3}{4}\right]$	[8,22]
9	$\left[x+\frac{1}{2},\ x+\frac{1}{2},\ \frac{3}{4}\right]$	[9,31]
10	$\left[\frac{1}{2}-x,\ x+\frac{1}{2},\ \frac{3}{4}\right]$	[10,29]
11	$\left[x+\frac{1}{2},\ \frac{1}{2}-x,\ \frac{3}{4}\right]$	[11,28]
12	$\left[x+\frac{1}{2},\ \frac{1}{2}-x,\ \frac{1}{4}\right]$	[12,27]
13	$\left[\frac{1}{2}-x,\ x+\frac{1}{2},\ \frac{1}{4}\right]$	[13,26]
14	$\left[\frac{1}{2}-x,\ \frac{1}{2}-x,\ \frac{3}{4}\right]$	[14,32]
15	$\left[x+\frac{1}{2},\ x+\frac{1}{2},\ \frac{1}{4}\right]$	[15,25]
16	$\left[\frac{1}{2}-x,\ \frac{1}{2}-x,\ \frac{1}{4}\right]$	[16,30]

Table 15: Wyckoff site: 16o, site symmetry: ..2'

No.	position	mapping
1	$\left[x,\ x+\frac{1}{2},\ 0\right]$	[1,31]
2	$\left[\frac{1}{2}-x,\ x,\ 0\right]$	[2,29]
3	$\left[x+\frac{1}{2},\ -x,\ 0\right]$	[3,28]
4	$\left[x,\ \frac{1}{2}-x,\ 0\right]$	[4,27]
5	$\left[-x,\ x+\frac{1}{2},\ 0\right]$	[5,26]
6	$\left[-x,\ \frac{1}{2}-x,\ 0\right]$	[6,32]
7	$\left[x+\frac{1}{2},\ x,\ 0\right]$	[7,25]
8	$\left[\frac{1}{2}-x,\ -x,\ 0\right]$	[8,30]
9	$\left[x+\frac{1}{2},\ x,\ \frac{1}{2}\right]$	[9,23]
10	$\left[-x,\ x+\frac{1}{2},\ \frac{1}{2}\right]$	[10,21]
11	$\left[x,\ \frac{1}{2}-x,\ \frac{1}{2}\right]$	[11,20]
12	$\left[x+\frac{1}{2},\ -x,\ \frac{1}{2}\right]$	[12,19]
13	$\left[\frac{1}{2}-x,\ x,\ \frac{1}{2}\right]$	[13,18]
14	$\left[\frac{1}{2}-x,\ -x,\ \frac{1}{2}\right]$	[14,24]
15	$\left[x,\ x+\frac{1}{2},\ \frac{1}{2}\right]$	[15,17]

continued ...

Table 15

No.	position	mapping
16	$\left[-x,\ \frac{1}{2}-x,\ \frac{1}{2}\right]$	[16,22]

Table 16: Wyckoff site: 32p, site symmetry: 1

No.	position	mapping
1	$[x,\ y,\ z]$	[1]
2	$[-y,\ x,\ z]$	[2]
3	$[y,\ -x,\ z]$	[3]
4	$[x,\ -y,\ -z]$	[4]
5	$[-x,\ y,\ -z]$	[5]
6	$[-x,\ -y,\ z]$	[6]
7	$[y,\ x,\ -z]$	[7]
8	$[-y,\ -x,\ -z]$	[8]
9	$\left[x+\frac{1}{2},\ y+\frac{1}{2},\ z+\frac{1}{2}\right]$	[9]
10	$\left[\frac{1}{2}-y,\ x+\frac{1}{2},\ z+\frac{1}{2}\right]$	[10]
11	$\left[y+\frac{1}{2},\ \frac{1}{2}-x,\ z+\frac{1}{2}\right]$	[11]
12	$\left[x+\frac{1}{2},\ \frac{1}{2}-y,\ \frac{1}{2}-z\right]$	[12]
13	$\left[\frac{1}{2}-x,\ y+\frac{1}{2},\ \frac{1}{2}-z\right]$	[13]
14	$\left[\frac{1}{2}-x,\ \frac{1}{2}-y,\ z+\frac{1}{2}\right]$	[14]
15	$\left[y+\frac{1}{2},\ x+\frac{1}{2},\ \frac{1}{2}-z\right]$	[15]
16	$\left[\frac{1}{2}-y,\ \frac{1}{2}-x,\ \frac{1}{2}-z\right]$	[16]
17	$\left[x,\ y,\ z+\frac{1}{2}\right]$	[17]
18	$\left[-y,\ x,\ z+\frac{1}{2}\right]$	[18]
19	$\left[y,\ -x,\ z+\frac{1}{2}\right]$	[19]
20	$\left[x,\ -y,\ \frac{1}{2}-z\right]$	[20]
21	$\left[-x,\ y,\ \frac{1}{2}-z\right]$	[21]
22	$\left[-x,\ -y,\ z+\frac{1}{2}\right]$	[22]
23	$\left[y,\ x,\ \frac{1}{2}-z\right]$	[23]
24	$\left[-y,\ -x,\ \frac{1}{2}-z\right]$	[24]
25	$\left[x+\frac{1}{2},\ y+\frac{1}{2},\ z\right]$	[25]
26	$\left[\frac{1}{2}-y,\ x+\frac{1}{2},\ z\right]$	[26]
27	$\left[y+\frac{1}{2},\ \frac{1}{2}-x,\ z\right]$	[27]
28	$\left[x+\frac{1}{2},\ \frac{1}{2}-y,\ -z\right]$	[28]
29	$\left[\frac{1}{2}-x,\ y+\frac{1}{2},\ -z\right]$	[29]
30	$\left[\frac{1}{2}-x,\ \frac{1}{2}-y,\ z\right]$	[30]
31	$\left[y+\frac{1}{2},\ x+\frac{1}{2},\ -z\right]$	[31]
32	$\left[\frac{1}{2}-y,\ \frac{1}{2}-x,\ -z\right]$	[32]